BASH SCRIPTING

1. Check your Shell

```
[root@ip-172-31-35-54 scripts]# echo $0
bash
[root@ip-172-31-35-54 scripts]#
```

2. Created test file
   used vi editor to edit
   #!/bin/bash
   a='viquar'
   echo $a

```
root@ip-172-31-35-54:/home/ec2-user/viquar/scripts
[root@ip-172-31-35-54 scripts]# pwd
/home/ec2-user/viquar/scripts
[root@ip-172-31-35-54 scripts]# ls
test
[root@ip-172-31-35-54 scripts]# chmod 777 test
[root@ip-172-31-35-54 scripts]# cat test
#!/bin/bash
# Purpose: testing script format
# date: 15/10/2026
# Modification: 15/10/2026
a='Viquar'
echo $a
[root@ip-172-31-35-54 scripts]# ./test
Viquar
[root@ip-172-31-35-54 scripts]#
```

3. Executed Basic sequence scripting

```
[root@ip-172-31-35-54 scripts]# vi basic.bash
[root@ip-172-31-35-54 scripts]# cat basic.bash
#!/bin/bash
pwd
ls
whoami
cal
touch abc 123 xyz
[root@ip-172-31-35-54 scripts]# ./basic.bash
bash: ./basic.bash: Permission denied
[root@ip-172-31-35-54 scripts]# chmod 755 basic.bash
[root@ip-172-31-35-54 scripts]# ./basic.bash
/home/ec2-user/viquar/scripts
basic.bash  test
root
    October 2026
Su Mo Tu We Th Fr Sa
             1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

[root@ip-172-31-35-54 scripts]# ls
123  abc  basic.bash  test  xyz
[root@ip-172-31-35-54 scripts]#
```

4. Executed Basic sequence scripting

```
root@ip-172-31-35-54:/home/ec2-user/viquar/scripts
[root@ip-172-31-35-54 scripts]# chmod 755 admin.bash
[root@ip-172-31-35-54 scripts]# cat admin.bash
#!/bin/bash
top -10
df -h
free -m
uptime
iostat
[root@ip-172-31-35-54 scripts]# ./admin.bash
top: unknown option '0'
Usage:
  top -hv | -bcEeHiOSs1 -d secs -n max -u|U user -p pid(s) -o field -w [cols]
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          4.0M     0  4.0M   0% /dev
tmpfs             457M     0  457M   0% /dev/shm
tmpfs             183M  444K  183M   1% /run
efivarfs          128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1    8.0G  1.7G  6.3G  21% /
tmpfs             457M     0  457M   0% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs              92M     0   92M   0% /run/user/1000
               total        used        free      shared  buff/cache   available
Mem:             913         170         454           0         287         610
Swap:              0           0           0
 10:07:24 up 34 min,  2 users,  load average: 0.03, 0.01, 0.00
Linux 6.18.54-121.166.amzn2023.x86_64 (ip-172-31-35-54.ap-southeast-2.compute.internal)
)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.85    0.00    0.15    0.06    0.07   98.87

Device             tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
nvme0n1           3.82        81.00       140.68         0.00     165931     288180          0


[root@ip-172-31-35-54 scripts]#
```

5. Executed Define scripting

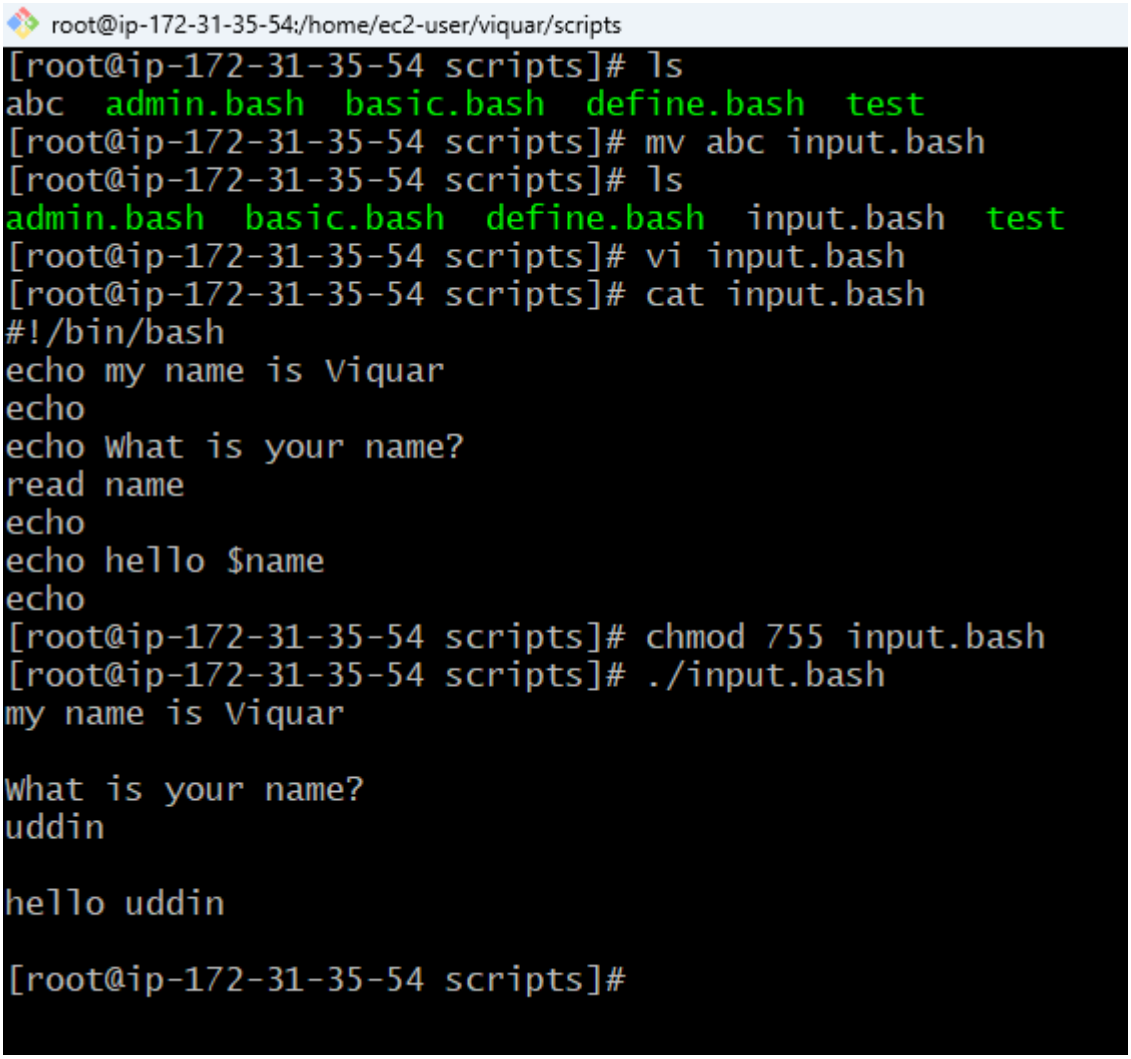
```
[root@ip-172-31-35-54 scripts]# vi define.bash
[root@ip-172-31-35-54 scripts]# chmod 755 define.bash
[root@ip-172-31-35-54 scripts]# ./define.bash
run varibles tasks

/home/ec2-user/viquar/scripts
abc  admin.bash  basic.bash  define.bash  test
root
Thu Oct 15 10:14:13 UTC 2026
    October 2026
Su Mo Tu We Th Fr Sa
              1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

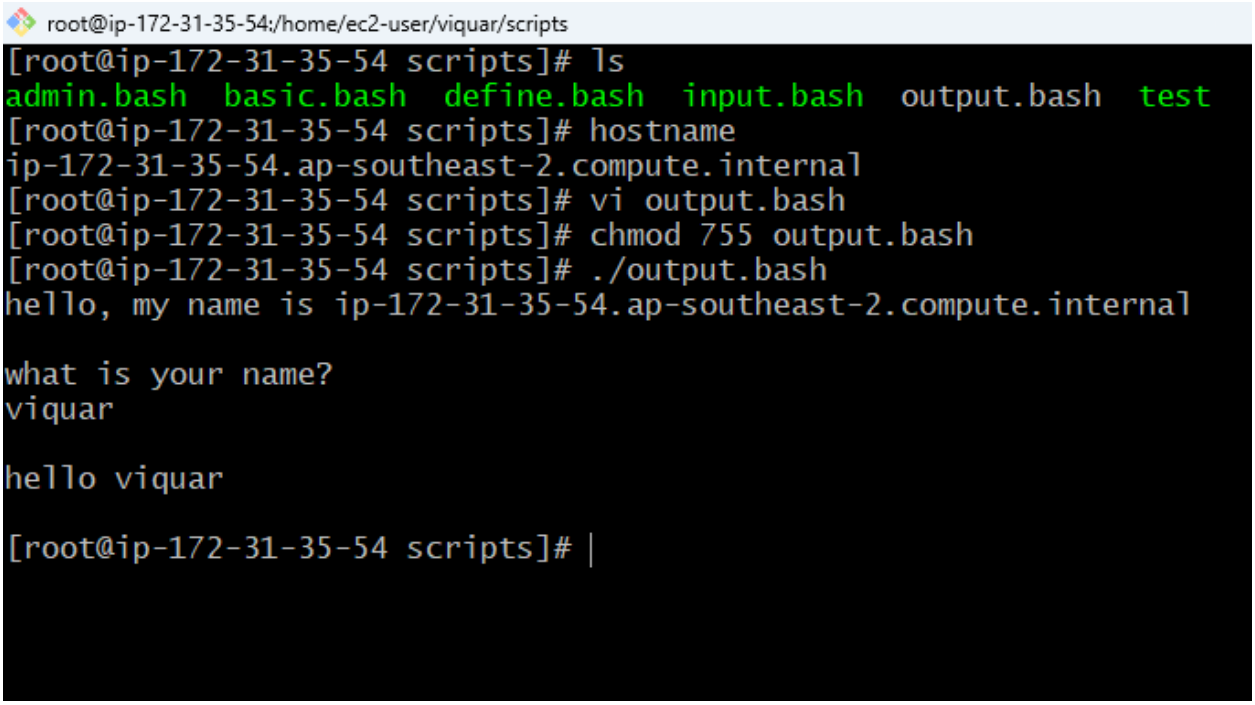
[root@ip-172-31-35-54 scripts]# cat define.bash
#!/bin/bash
#Define Variables
p=pwd
l=ls
wi=whoami
d=date
c=cal

echo run varibles tasks
echo
$p
$l
$wi
$d
$c
[root@ip-172-31-35-54 scripts]#
```

6. Executed input scripting



7. Executed Output scripting



8. Executed Condition scripting

```
[root@ip-172-31-35-54 scripts]# cat condition.bash
#!/bin/bash
count=100
if [ $count -eq 100]
then
        echo count is 100
else
        echo count is not 100
fi
[root@ip-172-31-35-54 scripts]# vi condition.bash
[root@ip-172-31-35-54 scripts]# ./condition.bash
./condition.bash: line 3: [: missing `]'
count is not 100
[root@ip-172-31-35-54 scripts]# vi condition.bash
[root@ip-172-31-35-54 scripts]# ./condition.bash
count is 100
[root@ip-172-31-35-54 scripts]#
```

9. Executed if -else statement scripting

root@ip-172-31-35-54:/home/ec2-user/viquar/scripts

```
[root@ip-172-31-35-54 scripts]# ls
admin.bash  basic.bash  condition.bash  define.bash  else.bash
[root@ip-172-31-35-54 scripts]# cat else.bash
#!/bin/bash
if [ -e /home/error.txt]
then
echo "File exist"
else
echo "File does not exist"
fi

[root@ip-172-31-35-54 scripts]# ./else.bash
./else.bash: line 2: [: missing `]'
File does not exist
[root@ip-172-31-35-54 scripts]#
```

10. Executed Case statement scripting

```
[root@ip-172-31-35-54 scripts]# cat case.bash
#!/bin/bash

echo
echo Please chose one of the options below
echo
echo 'a = Display Date and Time'
echo 'b = List file and directories'
echo 'c = List users logged in'
echo 'd = Check System uptime'
echo

        read choices

        case $choices in

a) date;;
b) ls;;
c) who;;
d) uptime;;
*) echo Invalid choice - Bye.
esac
[root@ip-172-31-35-54 scripts]# ./case.bash

Please chose one of the options below

a = Display Date and Time
b = List file and directories
c = List users logged in
d = Check System uptime

c
ec2-user pts/0        2026-10-15 09:36 (103.203.172.230)
```

11. Executed Loop statement scripting

```
See Rehan gone
See Rehan absent
See Rehan eat
See Rehan repaet
[root@ip-172-31-35-54 scripts]#
```

12. Executed Loop statement scripting

```
[root@ip-172-31-35-54 scripts]# vi loop.bash
[root@ip-172-31-35-54 scripts]# chmod loop.bash
chmod: missing operand after ‘loop.bash’
Try 'chmod --help' for more information.
[root@ip-172-31-35-54 scripts]# chmod 755 loop.bash
[root@ip-172-31-35-54 scripts]# ./loop.bash
hello 2 times
hello 4 times
hello 6 times
hello 8 times
hello 10 times
[root@ip-172-31-35-54 scripts]#
```

13. Executed Do-while scripting

```
[root@ip-172-31-35-54 scripts]# ./do-while.bash
10 seconds left to stop this process

9 seconds left to stop this process

8 seconds left to stop this process

7 seconds left to stop this process

6 seconds left to stop this process

5 seconds left to stop this process

4 seconds left to stop this process

3 seconds left to stop this process

2 seconds left to stop this process

1 seconds left to stop this process

process is stopped!!!
[root@ip-172-31-35-54 scripts]#
```

14. Executed Exit Status scripting

```
[root@ip-172-31-35-54 scripts]# vi test2.bash
[root@ip-172-31-35-54 scripts]# ./test2.bash
hello
0
[root@ip-172-31-35-54 scripts]# vi test2.bash
[root@ip-172-31-35-54 scripts]# ./test2.bash
./test2.bash: line 3: echi: command not found
127
```

15. Executed Exit Status scripting

```
[root@ip-172-31-35-54 scripts]# ./ test1.bash
bash: ./: Is a directory
[root@ip-172-31-35-54 scripts]# cat test1.bash
#!/bin/bash

echo hello
echo $?
[root@ip-172-31-35-54 scripts]# ./test1.bash
hello
0
```

16. Executed exit scripting

root@ip-172-31-35-54:/home/ec2-user/viquar/scripts

```
[root@ip-172-31-35-54 scripts]# cat test3.bash
#!/bin/bash

ls -l /home/ec2-user/viquar/scripts/empty

if [[ $? -eq 1 ]]
then
echo File exist
else
echo File does not exist
fi
[root@ip-172-31-35-54 scripts]# ./test3.bash
total 0
File does not exist
[root@ip-172-31-35-54 scripts]#
```